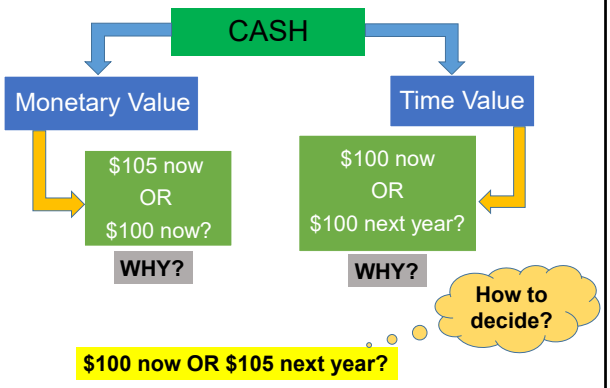
F78AB Actuarial and Financial Mathematics

Chapter 2: Interest and Present Value

Properties of Cash



2.1 The Time Value of Money

- Would you rather have £100 now or next week? Why?
  - **Deferred use** – you do not have use of the money.
  - **Risk** – the money might never be paid to you.
  - **Inflation** – prices might rise.

So, you expect a reward for letting someone else have use of your money for a period of time – this is known as **interest**.

E.g. 2.1.1: Accumulated Value

- You deposit £1,000 in the savings account of Bank ABC for one year.
- The bank pays **interest at 3% per annum**.
- In one year's time you will **receive £1,030**, which consists of the return of your **capital** (£1,000) and **interest** (£30).

E.g. 2.1.2: Interest rate calculation

- You deposit £1,000 in the savings account of Bank ABC for one year.
- In one year's time you **receive £1,050**.
- This £1,050 consists of the return of your **capital** (£1,000) and **interest** (£50).
- You can deduce that the bank paid you **interest at the rate of 5% per annum**.

2.2 Effective Rate of Interest

- The **effective rate of interest per unit of time** is the amount of interest received at the end of one time unit per £1 (or other currency) invested at the start of that time unit.
- The effective rate of interest per time unit is most often denoted i.
- Other letters used are j, or subscripted symbols like i_1 and i_2.

Simple vs Compound Interest

- We know that £1 invested for one time unit at the effective interest rate of i per time unit becomes $(1+i)$.
- How much will you have if you invested £1 for **two, or more, time units** at an effective rate of interest i per time unit?



Simple vs Compound Interest

- Say you invest **£1000** into a bank account that pays interest at a rate of **12% per annum effective**.
- You decide to just leave the money there, i.e. no additional deposits and no withdrawals.
- How much will the account have at the end of 1 year?
- How much will the account have at the end of 2 years?

cont.

Simple vs Compound Interest

- How much will the account have at the end of
  - year 3,
  - year 4,
  - . . . ,
  - year 10?

It depends on how interest is calculated.

- Simple Interest, or Compound Interest.

Simple vs Compound Interest

- **Simple interest**
  - Interest is earned only on the initial capital deposited.
  - **Interest does not earn interest.**
  - Suppose an amount £1 is deposited into an account that pays **simple interest** at the rate of i per time unit. At the end of n time units the deposit would become

Simple vs Compound Interest

- **Compound interest**
  - Interest is earned on the initial capital deposited and from the interest that has been paid into the account.
  - **Interest earns interest.**
  - Suppose an amount £1 is deposited into an account that pays **compound interest** at the rate of i per time unit. At the end of n time units the deposit would become

WE WILL ASSUME COMPOUND INTEREST IN THIS COURSE UNLESS OTHERWISE STATED.

2.3 More than One Time Unit

- E.g. 2.3.1:
  - Invest £50 for 7 years at 4% per annum effective.
- E.g. 2.3.2 :
  - Invest £100 for 9 months at 2% per quarter year effective.
- E.g. 2.3.3 :
  - Invest £80 for 1 year at 6.25% per half year effective.

Round your final answers to 2 decimal places as these are monetary amounts.

2.4 Compounding over any number of time units

The compound interest formula $(1+i)^t$ extends to any real number $t \geq 0$.

- E.g. 2.4.1
  - Invest £2,000 for 4 months at 10.5% per annum effective.
- E.g. 2.4.2
  - Invest £2,500 for 1 month at 2% per quarter year effective.
- E.g. 2.4.3
  - Invest £100 for 3.6 years at 4.1% per half year effective.

2.5 Changing the Time Unit

- A contract may have interest rates and payments that have different time units.
- For example, a mortgage is a contract that has monthly cashflows but annual interest.
- It would be more convenient to work with an effective rate of interest per month.
- Suppose we have two time units denoted t_i years and t_j years.

2.5 Changing the Time Unit

- Let the effective rate of interest per t_i years be denoted i, and let the effective rate of interest per t_j years be denoted j.
- Using time unit t_i years, after 1 year we have
- Using time unit t_j years, after 1 year we have

2.5 Changing the Time Unit

- The two effective rates of interest are defined to be equivalent if we have the same amount of money after one year.
- Hence after 1 year we have,

2.5 Changing the Time Unit

- E.g. 2.5.1
  - What effective rate of interest per half-year is equivalent to 7% per annum effective?
- E.g. 2.5.2
  - What effective rate of interest per quarter year is equivalent to 15% per two years effective?
- E.g. 2.5.3
  - What effective rate of interest per 1.7 years is equivalent to 1.3% per month effective?

Comment on Rounding

Read
"Comment on Rounding"
on page 22
of the notes.

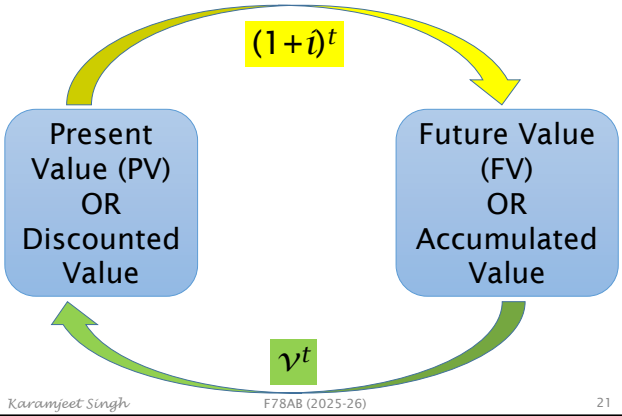
2.6 Accumulating and Discounting

- We say that money accumulates with interest. The amount to which money accumulates (with the addition of interest) is its accumulation or accumulated value (AV or FV).
- If i is the effective rate of interest per time unit, we call $(1+i)^t$ the accumulation factor over t time units.
- Money required now to meet a known future payment is the discounted value or present value (PV) of that payment.

2.6 Accumulating and Discounting

- If i is the effective rate of interest per time unit, we call $(1+i)^{-1}$ the discount factor per time unit.
- **Definition:** $v = (1+i)^{-1}$
- We tend to write v^t for the discount factor over t time units.
- Accumulating and Discounting are inverse operations.

2.6 Accumulating and Discounting



2.6 Accumulating and Discounting

- E.g. 2.6.1*
  - Joseph needs £2,650 in 1 years' time. The bank offers interest at 6% per annum. How much must he deposit now to meet his needs?

2.6 Accumulating and Discounting

- E.g. 2.6.2*
  - Albert needs £20,000 in 5 years' time, how much must he deposit now into a bank account that pays interest at 2.9186% per half-year effective?

2.6 Accumulating and Discounting

- E.g. 2.6.4
  - Let $v = 0.94$ be the discount factor per half-year.
    i) What is the effective rate of interest per half-year?
    ii) What is the equivalent discount factor per year?
    Give your answers to 6 decimal places.

2.7 The Effective Rate of Discount

- **Definition:** Let i be the effective rate of interest per time unit, so that $v = (1+i)^{-1}$ is the corresponding discount factor. The effective rate of discount per unit of time, d, is defined as $d = 1 - v$.
- Hence, for each of the operations of accumulating and discounting, we have an effective rate and a corresponding factor

Operation	Rate	Factor
Accumulation	i	$1+i$
Discounting	d	$1-d = v$

Useful Results

	Value of		
In terms of	i	v	d
i	-	$\frac{1}{1+i}$	$\frac{i}{1+i}$
v	$\frac{1}{v} - 1$	-	$1 - v$
d	$\frac{d}{1-d}$	$1 - d$	-

2.7 The Effective Rate of Discount

- E.g. 2.7.1
  - Given $i = 0.04$ effective per time unit, find v and d, correct to 6 decimal places.

2.8 The Value at Time s of a Cashflow at Time t

Say we have a cashflow X at time t and that the effective rate of interest is i per time unit.

- **Case 1: $s \geq t$**
  - We must **accumulate** the cashflow from time t to time s,
- **Case 2: $s < t$**
  - We must **discount** the cashflow from time t to time s,

E.g. 2.8: Assume the effective rate of interest is 5% per half-year.

- E.g. 2.8.1(i)
  - What is the value at time $s = 3$ years of a cashflow of £1,000 payable at time $t = -2$ years?
- E.g. 2.8.1(ii)
  - What is the value at time $s = 4$ years of a cashflow of £500 due at time $t = 7.75$ years?

2.9 Non-Constant Interest Rates

Interest rates are not always constant. We can use the same theory as earlier to determine present values and accumulations if we know the interest rate over different periods of time.

- E.g. 2.9.1
  - Let the effective interest rate be 4% per annum during 2023 and 6% per annum during 2024. What is the accumulated value as at 31/12/24 of £1,000 invested on 1/1/23?

2.9 Non-Constant Interest Rates

- E.g. 2.9.2

The effective rate of interest will be 5% per annum during 2023, 5.5% per annum during 2024 and 6% per annum during 2025. Calculate the accumulation of each of the following cashflows.

i. £1,000 invested on 1/1/23 for 3 years?
ii. £1,000 invested on 1/7/23 for 2 years?
iii. £1,000 invested on 1/10/24 for 1 year?

2.9 Non-Constant Interest Rates

In general suppose that an amount of X is invested between time t_o and t_n with effective interest rate being i_r between times t_{r-1} and t_r then the accumulated value of X at time t_n is

$$X\,(1+i_1)^{t_1-t_0}\,(1+i_2)^{t_2-t_1}\,\ldots\,(1+i_n)^{t_n-t_{n-1}} = X\prod_{r=1}^{n}(1+i_r)^{t_r-t_{r-1}}$$

2.9 Non-Constant Interest Rates

Alternatively suppose we require the Present Value (PV) at time t_o of X due at time t_n. Then this PV is

$$X\,(1+i_1)^{t_0-t_1}\,(1+i_2)^{t_1-t_2}\,\ldots\,(1+i_n)^{t_{n-1}-t_n} = X\prod_{r=1}^{n}(1+i_r)^{t_{r-1}-t_r}$$

2.9 Non-Constant Interest Rates

• E.g. 2.9.3

Starting now, the effective rate of interest will be 3% per annum for 6 months, 4% per annum for the following one year, and 5% per annum for the following 2 years. What is the accumulation factor over the period [0, 3.5] (measuring time in years)?

2.9 Non-Constant Interest Rates

• E.g. 2.9.4

Starting now, the effective rate of interest will be 2% per half year for 1 year, 1.5% per quarter year for 6 months, and 8% p.a. for the following 18 months. What is the PV now of £1,000 due in 3 years' time?

2.10 General Accumulation and Discount Factors

It is sometimes convenient to work with a more general and abstract model of accumulation and discounting.

Definition

For arbitrary times t and s, with $t \le s$, define

- $V(t, s)$ = present value at time t of 1 payable at time s.
- $V(s, t) = V(t, s)^{-1}$ = accumulation at time s of 1 payable at time t.

2.10 General Accumulation and Discount Factors

Rules

- $V(t, s) > 0$ for any t and s.
- $V(t, s) = V(t, r)V(r, s)$ for any t, s and r.

Question

Show that $V(t, t) = 1$.

Next Chapter

Chapter 3: Cashflows and Annuities